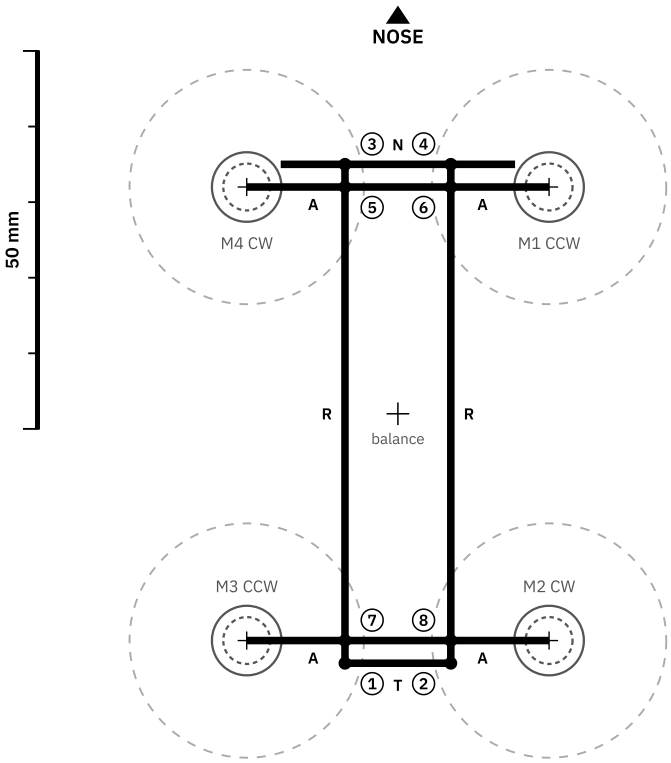
XS drone frame jig

Print on US Letter at Actual size (100 %), never Fit to page.
Both bars must read 50 mm on a caliper before you cut or lay anything.

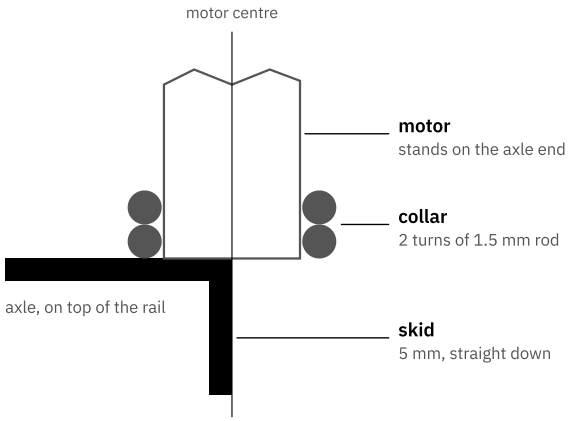
50 mm



Top view, 1:1. Black is rod, laid where it lies. Dots are joints in solder order.
Grey rings are the collars, dashed circles the 31 mm propellers.

Axle end, side view, 3 × size

Not for laying rod: the 1:1 view is on the left.



Collar check, 1:1



inside 6.2 mm, outside 9.2 mm
A wound collar should cover the grey ring.

Joints

Laps: each cross piece lies on top of the rails.
Collars sit on the axle ends; skids hang under them.
The battery straps between the rails, nose to tail;
the frame stays electrically floating.

Cut list, 1:1. Lay the rod on its bar and mark it at the end tick.

R rails · 2 × 66 mm · 1 mm rod

A axles · 2 × 40 mm · 1 mm rod

N nose bar · 1 × 31 mm · 1 mm rod

T tail bar · 1 × 14 mm · 1 mm rod

S skids · 4 × 5 mm · 1 mm rod

C collar blanks · 4 × 60 mm · 1.5 mm rod

All ten 1 mm pieces come from one 300 mm rod: 277 mm plus the saw cuts. The collars take two 60 mm blanks from each of two 150 mm coupons.

Build order

1. Test first. On two 1.5 mm coupons, wind one collar and solder one lap and one T with the stainless flux. Bend them hard. A joint that pops off clean means more heat and more flux before the frame.
2. Cut. Lay each rod on its bar below, mark the end tick, saw 0.5 mm long and sand square to the mark.
3. Set up. Tape this sheet to the silicone mat (never the cutting mat). Put a Kapton square over each numbered dot, then hold the rails on their lines with Kapton strips.
4. Solder 1 to 4: tail bar, then nose bar, both laid across the tops of the rails. Check the rail diagonals: 67.5 mm both ways.
5. Solder 5 to 8: both axles across the tops of the rails, each end on its motor centre.
6. Collars. Set one on each axle end, lower turn on the axle, with the 6 mm former left inside so it stays round. Solder, then pull the former out. A motor should push in by hand.
7. Flip the frame and solder a 5 mm skid under each axle end, straight down.
8. Wash off every trace of flux (warm water, a pinch of baking soda), dry and weigh: about 4.4 g.